Operational Manual: Homogeneous-Background Transition-Matrix Extraction and Meta-Atom Scattering

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Abstract

This manual specifies a reproducible full-wave workflow for extracting the electromagnetic transition matrix of a finite meta-atom made from realistic, possibly dispersive and lossy materials. The meta-atom is embedded in one homogeneous, isotropic, lossless background medium, and the numerical domain is terminated by perfectly matched layers (PMLs) on all six sides. Incident and scattered fields are expanded in regular and outgoing vector spherical wave functions (VSWFs), respectively. Multiple meta-atoms are then coupled through homogeneous-medium translation operators. Substrates, planar interfaces, waveguides, literal atomic-scale quantum scatterers, nonlinear materials, and particles intersecting the PML are outside the scope of this edition.

1 Scope, assumptions, and deliverables

The procedure is valid under the following assumptions.

1. Each meta-atom is finite and spatially localized.
2. Its constitutive response is linear and time invariant at each sampled frequency. Electric, magnetic, anisotropic, bianisotropic, dispersive, and lossy material models are permitted when they are supported consistently by the full-wave solver.
3. Outside the meta-atom, the physical region is a homogeneous, isotropic, lossless background with permittivity ϵ_b, permeability μ_b, wave number $k_b = \omega\sqrt{\epsilon_b \mu_b}$, and impedance $Z_b = \sqrt{\mu_b/\epsilon_b}$.
4. No substrate, interface, periodic boundary, or waveguide is present during isolated extraction. PMLs approximate radiation into an unbounded background; they are not part of the physical scatterer.
5. The expansion origin, coordinate axes, time convention, material models, and VSWF normalization are fixed and stored with every output.

The workflow produces a frequency-dependent matrix $\mathbf{T}(\omega)$ that maps incident regular VSWF coefficients to outgoing scattered VSWF coefficients. The principal deliverables are listed in Table 1.

Table 1: Computational stages and required outputs.

Stage	Operation	Required inputs	Required outputs
1. Full-wave extraction	PML-terminated simulations	Geometry; material models; background; frequency grid; plane-wave directions and polarizations; mesh and PML settings	Complex total or scattered $\mathbf{E}, \mathbf{H}$ fields; incident-field metadata; convergence metadata
2. VSWF projection	Weighted spherical surface projection	Field data; expansion origin; $L_{\max}$; monitor radius; quadrature; VSWF convention	Incident matrix $\mathbf{A}$; scattered matrix $\mathbf{F}$; projection residuals
3. T-matrix recovery	Rank-revealing weighted least squares	$\mathbf{A}, \mathbf{F}$; illumination weights; SVD cutoff	$\mathbf{T}(\omega)$; singular values; rank; fit and holdout errors
4A. Finite cluster	Homogeneous-medium Foldy-Lax solve	Meta-atom T-matrices; positions; orientations; incident field	Coupled coefficients; angular far field; cross sections or specified mode overlaps
4B. Periodic array (optional)	Bloch lattice aggregation	Unit-cell T-matrices; lattice; Bloch wave vector; converged lattice sum	Power-normalized Floquet reflection and transmission channels

Terminology. The object extracted with PML boundaries is called a *localized meta-atom* or *scatterer*, not a periodic unit cell. The transition matrix is distinct from the transfer matrix used for one-dimensional stratified media.

2 Fixed electromagnetic and VSWF conventions

All phasors use the time dependence $\exp(-i\omega t)$. Outgoing spherical waves therefore use the spherical Hankel function of the first kind. Coordinates are right handed, θ is measured from $+z$, and ϕ is measured from $+x$ toward $+y$. For scalar passive constitutive parameters, this convention gives $\operatorname{Im}\epsilon \geq 0$ and $\operatorname{Im}\mu \geq 0$. Scalar spherical harmonics are orthonormal,

$$\int_{4\pi} Y_{lm}^*(\hat{\boldsymbol{r}}) Y_{l'm'}(\hat{\boldsymbol{r}})\, d\Omega = \delta_{ll'}\delta_{mm'}, \tag{1}$$

and use the Condon-Shortley phase. Define

$$\boldsymbol{X}_{lm}(\hat{\boldsymbol{r}}) = \frac{\boldsymbol{L} Y_{lm}(\hat{\boldsymbol{r}})}{\sqrt{l(l+1)}}, \qquad \boldsymbol{L} = -i\boldsymbol{r}\times\nabla. \tag{2}$$

The regular ($p = 1$) and outgoing ($p = 3$) VSWFs are

$$\mathbf{M}_{lm}^{(p)}(k_b\boldsymbol{r}) = z_l^{(p)}(k_b r)\boldsymbol{X}_{lm}(\hat{\boldsymbol{r}}), \tag{3}$$

$$\mathbf{N}_{lm}^{(p)}(k_b\boldsymbol{r}) = \frac{1}{k_b}\nabla\times\mathbf{M}_{lm}^{(p)}(k_b\boldsymbol{r}), \tag{4}$$

where $z_l^{(1)} = j_l$ and $z_l^{(3)} = h_l^{(1)}$.

With the compound mode index $\alpha = (\tau, l, m)$, where $\tau \in \{M, N\}$, $l = 1, \ldots, L_{\max}$, and $m = -l, \ldots, l$, the electric fields are

$$\mathbf{E}_{\text{inc}}(\boldsymbol{r}) = E_0 \sum_{l=1}^{L_{\max}} \sum_{m=-l}^{l} \left[a_{Mlm}\mathbf{M}_{lm}^{(1)} + a_{Nlm}\mathbf{N}_{lm}^{(1)}\right], \tag{5}$$

$$\mathbf{E}_{\text{sca}}(\boldsymbol{r}) = E_0 \sum_{l=1}^{L_{\max}} \sum_{m=-l}^{l} \left[f_{Mlm}\mathbf{M}_{lm}^{(3)} + f_{Nlm}\mathbf{N}_{lm}^{(3)}\right]. \tag{6}$$

The corresponding magnetic fields follow from Maxwell's equations:

$$\mathbf{H}_{\text{inc}} = -\frac{iE_0}{Z_b}\sum_{lm}\left[a_{Mlm}\mathbf{N}_{lm}^{(1)} + a_{Nlm}\mathbf{M}_{lm}^{(1)}\right], \tag{7}$$

$$\mathbf{H}_{\text{sca}} = -\frac{iE_0}{Z_b}\sum_{lm}\left[f_{Mlm}\mathbf{N}_{lm}^{(3)} + f_{Nlm}\mathbf{M}_{lm}^{(3)}\right]. \tag{8}$$

Every implementation shall serialize the explicit ordered list of compound mode indices rather than relying on an undocumented array order.

The number of retained electric and magnetic modes is

$$D = 2\sum_{l=1}^{L_{\max}}(2l+1) = 2L_{\max}(L_{\max}+2). \tag{9}$$

For a fixed frequency, origin, orientation, background, and basis convention, the transition matrix is defined by

$$\boxed{\boldsymbol{f} = \mathbf{T}\boldsymbol{a}.} \tag{10}$$

It is independent of the incident field profile only for a linear system. It still depends on frequency, all constitutive tensors, background medium, coordinate origin and orientation, and basis normalization.

3 Stage 1: PML-terminated full-wave extraction

3.1 Geometry, monitor, and PML placement

Choose an expansion origin $\boldsymbol{r}_0$, preferably near the center of the minimum circumscribing sphere. Define

$$a = \max_{\boldsymbol{r}\in\text{object}} \|\boldsymbol{r} - \boldsymbol{r}_0\|, \qquad d_{\text{PML}} = \min_{s\in\text{six faces}} d(\boldsymbol{r}_0, \text{inner PML face } s). \tag{11}$$

The spherical monitor radius must satisfy

$$\boxed{a < r_{\text{mon}} < d_{\text{PML}}.} \tag{12}$$

The entire monitor must lie in the homogeneous physical region and outside the PML. Leave several local mesh cells between the material and monitor and between the monitor and the inner PML face. No universal PML distance guarantees accuracy; distance, thickness, and absorption grading must be verified by the convergence tests in Section 7.

The monitor is allowed to lie in the radiating near field. It need not be in the far field, because the outgoing VSWF expansion is valid everywhere outside a sphere enclosing all sources and material inhomogeneity.

3.2 Incident excitation set

Use a full-sphere set of plane-wave directions $\widehat{\boldsymbol{k}}_q$. A Lebedev grid is convenient because it distributes directions and supplies angular weights. For each direction, run two orthogonal transverse polarizations $\widehat{\boldsymbol{e}}_{q1}$ and $\widehat{\boldsymbol{e}}_{q2} = \widehat{\boldsymbol{k}}_q \times \widehat{\boldsymbol{e}}_{q1}$. The phase reference is the expansion origin $\boldsymbol{r}_0$:

$$\mathbf{E}_{\text{inc}}^{(q)}(\boldsymbol{r}) = E_0 \widehat{\boldsymbol{e}}_q e^{ik_b \widehat{\boldsymbol{k}}_q \cdot (\boldsymbol{r} - \boldsymbol{r}_0)}, \qquad \mathbf{H}_{\text{inc}}^{(q)} = Z_b^{-1} \widehat{\boldsymbol{k}}_q \times \mathbf{E}_{\text{inc}}^{(q)}. \tag{13}$$

Choose the first polarization by a deterministic reference-axis rule that remains nonsingular at the poles, and store both polarization vectors explicitly.

For K total polarization-resolved excitations, require

$$K \geq D, \qquad \text{rank}(\mathbf{A}) = D. \tag{14}$$

As a practical starting point, use $K \gtrsim 1.5D$ to $2D$ and then check singular values. The Lebedev algebraic degree $p \geq 2L_{\max}$ is useful for angular coverage, but it does not by itself guarantee rank, conditioning, or full-wave accuracy.

Do not use a raw hemisphere reduction in the first validated implementation. Mirror symmetry may later reduce cost only after the missing incident responses are reconstructed with the correct electric and magnetic multipole parity transformations and the reduced matrix is shown to retain full rank in each symmetry block.

3.3 Fields to export

For every excitation and frequency, obtain the complex scattered fields

$$\mathbf{E}_{\text{sca}} = \mathbf{E}_{\text{tot}} - \mathbf{E}_{\text{inc}}, \qquad \mathbf{H}_{\text{sca}} = \mathbf{H}_{\text{tot}} - \mathbf{H}_{\text{inc}}. \tag{15}$$

Use a solver's scattered-field formulation when its convention has been verified. Otherwise, subtract the analytical incident field or an empty-domain reference computed with identical mesh, source, boundaries, amplitude, origin, and phasor convention. Export both $\mathbf{E}$ and $\mathbf{H}$ whenever possible; their simultaneous projection provides a stronger consistency test than either field alone [5].

4 Stage 2: Spherical projection and T-matrix recovery

4.1 Weighted surface projection

Sample the monitor using Gauss-Legendre nodes in $\mu = \cos\theta$ and uniformly spaced nodes in ϕ. A conservative starting grid is

$$N_\theta \geq 2L_{\max} + 2, \qquad N_\phi \geq 4L_{\max} + 4, \tag{16}$$

followed by a quadrature-doubling convergence test. Interpolate the Cartesian solver fields to the sphere and rotate them into $(\widehat{\boldsymbol{\theta}}, \widehat{\boldsymbol{\phi}})$ components.

At every quadrature point s, form the scaled tangential field vector

$$\boldsymbol{u}_s = \begin{bmatrix} E_{\theta,s} & E_{\phi,s} & Z_b H_{\theta,s} & Z_b H_{\phi,s} \end{bmatrix}^T. \tag{17}$$

Let $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(3)}$ be the matrices whose columns contain these four tangential components for unit-amplitude regular and outgoing VSWFs, respectively, evaluated at r_{mon}. With the diagonal angular-weight matrix $\mathcal{W}$, extract coefficients by weighted least squares:

$$\boldsymbol{a} = \arg\min_{\boldsymbol{c}} \left\| \mathcal{W}^{1/2} \left(\mathcal{B}^{(1)} \boldsymbol{c} - \boldsymbol{u}_{\text{inc}} \right) \right\|_2, \tag{18}$$

$$\boldsymbol{f} = \arg\min_{\boldsymbol{c}} \left\| \mathcal{W}^{1/2} \left(\mathcal{B}^{(3)} \boldsymbol{c} - \boldsymbol{u}_{\text{sca}} \right) \right\|_2. \tag{19}$$

Use a rank-revealing QR or singular-value decomposition (SVD), not an unqualified normal-equation inverse. Record the projection residual

$$\epsilon_{\text{proj}} = \frac{\|\mathcal{W}^{1/2}(\mathcal{B}\boldsymbol{c} - \boldsymbol{u})\|_2}{\|\mathcal{W}^{1/2}\boldsymbol{u}\|_2}. \tag{20}$$

Apply the same projection to the analytical incident field and reconstruct it from the recovered $\boldsymbol{a}$. On an independent valid monitor sphere, require and report

$$\epsilon_{\text{inc}} = \frac{\|\mathcal{W}^{1/2}(\mathcal{B}^{(1)}\boldsymbol{a} - \boldsymbol{u}_{\text{inc}})\|_2}{\|\mathcal{W}^{1/2}\boldsymbol{u}_{\text{inc}}\|_2}. \tag{21}$$

This directly tests the origin, phase, polarization, and basis normalization used by both the full-wave model and the extraction code.

This numerical projection defines the coefficient normalization operationally. An analytical orthogonality projection may replace it only if all radial factors, phases, negative-m conventions, and electric/magnetic normalizations are derived consistently with Eqs. (5)-(6).

4.2 Recovery from multiple illuminations

Collect the coefficient vectors from all K excitations as

$$\mathbf{A} = \begin{bmatrix} \boldsymbol{a}^{(1)} & \cdots & \boldsymbol{a}^{(K)} \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} \boldsymbol{f}^{(1)} & \cdots & \boldsymbol{f}^{(K)} \end{bmatrix}. \tag{22}$$

Then $\mathbf{F} = \mathbf{T}\mathbf{A}$. The unweighted least-squares estimator is

$$\widehat{\mathbf{T}} = \mathbf{F}\mathbf{A}^{+}. \tag{23}$$

If the incident directions carry nonuniform weights collected in $\mathbf{W}_{\text{inc}}$, repeat each directional weight for both polarizations and define

$$\mathbf{A}_w = \mathbf{A}\mathbf{W}_{\text{inc}}^{1/2}, \qquad \mathbf{F}_w = \mathbf{F}\mathbf{W}_{\text{inc}}^{1/2}, \qquad \boxed{\widehat{\mathbf{T}} = \mathbf{F}_w\mathbf{A}_w^{+}}. \tag{24}$$

This multiple-plane-wave recovery requires the corrected SVD ordering described in Refs. [3, 4]. Report the singular spectrum of the weighted incident matrix and the effective numerical rank. A starting relative SVD cutoff of 10^{-10} to 10^{-8} may be tested in double precision, but the final cutoff must be selected from the observed singular-value gap and holdout performance.

The training consistency error is

$$\epsilon_{\text{fit}} = \frac{\|\mathbf{F} - \widehat{\mathbf{T}}\mathbf{A}\|_F}{\|\mathbf{F}\|_F}. \tag{25}$$

An independent set of unused plane waves must also satisfy

$$\epsilon_{\text{hold}} = \frac{\|\mathbf{F}_{\text{hold}} - \widehat{\mathbf{T}}\mathbf{A}_{\text{hold}}\|_F}{\|\mathbf{F}_{\text{hold}}\|_F}. \tag{26}$$

Use an independently rotated full-sphere direction set for this test, rather than withholding a symmetry-related subset of the fitting grid.

4.3 Multipole truncation

Let $x = k_b a$ be the size parameter. Wiscombe-type Mie truncation expressions may be used only to choose an initial $L_{\max}$; they are not rigorous bounds for arbitrary flat, sharp, high-index, or resonant meta-atoms [6]. Hold $L_{\max}$ fixed over a stored frequency sweep if a fixed matrix dimension is required, using the largest converged value. Confirm convergence by repeating the extraction at $L_{\max} + 1$ or $L_{\max} + 2$ and comparing the common low-order block and predicted observables.

5 Stage 3A: Finite-cluster aggregation

For N_p localized meta-atoms, let $\boldsymbol{a}_i$ be the total regular exciting coefficients at the origin of particle i, $\boldsymbol{a}_i^{\text{inc}}$ the externally imposed incident coefficients, and $\boldsymbol{f}_i$ its outgoing coefficients. The homogeneous-medium Foldy-Lax equations are

$$\boldsymbol{a}_i = \boldsymbol{a}_i^{\text{inc}} + \sum_{j \neq i} \mathbf{A}_{ij}^{(3\to1)} \boldsymbol{f}_j, \tag{27}$$

$$\boldsymbol{f}_i = \mathbf{T}_i \boldsymbol{a}_i. \tag{28}$$

Here $\mathbf{A}_{ij}^{(3\to1)}$ translates an outgoing VSWF centered at j into regular VSWFs centered at i. Rotations of a meta-atom are handled by rotating its T-matrix with Wigner matrices before assembly.

With block-diagonal $\mathbb{T} = \mathrm{diag}(\mathbf{T}_1, \ldots, \mathbf{T}_{N_p})$ and a block translation matrix $\mathbb{A}$ whose diagonal blocks vanish,

$$\boxed{(\mathbb{I} - \mathbb{T}\mathbb{A})\boldsymbol{f} = \mathbb{T}\boldsymbol{a}^{\mathrm{inc}}.} \tag{29}$$

For small systems, solve Eq. (29) directly. For large systems, use an iterative solver with residual monitoring and appropriate acceleration; the long-range interaction matrix is generally dense rather than sparse.

For a plane-wave drive, construct $\boldsymbol{a}_i^{\mathrm{inc}}$ at every particle origin with the corresponding phase factor $\exp[ik_b\widehat{\boldsymbol{k}} \cdot (\boldsymbol{r}_i - \boldsymbol{r}_{\mathrm{ref}})]$ relative to the stored global phase reference; do not copy one coefficient vector to every particle without this translation. Store the Euler-angle and Wigner-matrix conventions used for each rotated body-frame T-matrix.

5.1 Translation convergence condition

A practical sufficient condition for conventional outgoing-to-regular spherical translation is nonoverlap of the particles' circumscribing spheres:

$$\|\boldsymbol{r}_i - \boldsymbol{r}_j\| > a_i + a_j. \tag{30}$$

If this condition fails, increasing $L_{\max}$ does not necessarily repair the divergent local spherical expansion. Use plane-wave-mediated coupling, a verified distributed-source method, or group the close objects into a composite cluster T-matrix [7].

5.2 Finite-cluster observables

The total scattered field is the coherent sum of translated outgoing expansions:

$$\mathbf{E}_{\mathrm{sca}}(\boldsymbol{r}) = E_0 \sum_{i=1}^{N_p} \sum_{\alpha} f_{i\alpha} \boldsymbol{\Psi}_{\alpha}^{(3)}(k_b(\boldsymbol{r} - \boldsymbol{r}_i)) . \tag{31}$$

Use the far-field asymptotics of the same normalized VSWFs to calculate angular scattering. For a single incident plane wave in the homogeneous host,

$$\frac{d\sigma}{d\Omega} = \lim_{r\to\infty} r^2 \frac{\|\mathbf{E}_{\mathrm{sca}}(r\widehat{\boldsymbol{r}})\|^2}{\|\mathbf{E}_{\mathrm{inc}}\|^2}. \tag{32}$$

A finite cluster under an infinite plane wave does not automatically possess unique two-port S_{11} and S_{21}. If port-like quantities are required, specify finite input and output modes, reference planes, and power-normalized overlap integrals.

6 Stage 3B: Infinite periodic aggregation (optional)

For an infinite two-dimensional lattice in the same homogeneous background, use Bloch reduction rather than constructing an ever-larger finite matrix. For one scatterer per primitive cell, define the lattice interaction operator

$$\mathbb{W}(\boldsymbol{k}_\parallel) = \sum_{\boldsymbol{R}\neq\boldsymbol{0}} \mathbf{A}_{0,\boldsymbol{R}}^{(3\to1)} e^{i\boldsymbol{k}_\parallel\cdot\boldsymbol{R}}. \tag{33}$$

Here $\mathbf{A}_{0,\boldsymbol{R}}^{(3\to1)}$ maps an outgoing expansion centered in the cell at $\boldsymbol{R}$ into a regular expansion at the reference-cell origin, using the same translation definition as in Stage 3A. Then

$$\boldsymbol{f}_0 = \left[\mathbf{I} - \mathbf{T}\mathbb{W}(\boldsymbol{k}_\parallel)\right]^{-1} \mathbf{T}\boldsymbol{a}_0^{\mathrm{inc}}. \tag{34}$$

The lattice sum is conditionally convergent and shall be evaluated with a validated Ewald or reciprocal-space algorithm, with special convergence checks near Rayleigh anomalies [8].

Project the periodic scattered field into Floquet channels labeled by reciprocal vectors $\boldsymbol{G}$, with

$$\boldsymbol{k}_{\parallel,\boldsymbol{G}} = \boldsymbol{k}_\parallel + \boldsymbol{G}, \qquad k_{z,\boldsymbol{G}} = \sqrt{k_b^2 - \|\boldsymbol{k}_\parallel + \boldsymbol{G}\|^2}, \tag{35}$$

using $\mathrm{Im}\, k_{z,\boldsymbol{G}} \geq 0$. A channel is propagating when it carries nonzero real outward power. Report the power-normalized scattering matrix over all open diffraction orders and both polarizations. Scalar specular S_{11} and S_{21} are sufficient only when every nonzero order is evanescent.

7 Validation and acceptance criteria

Validation is part of the algorithm, not an optional post-processing step. Table 2 gives the minimum ladder. The numerical tolerances are recommended starting targets and may be tightened for the intended observable.

Table 2: Required validation ladder.

No.	Test	Acceptance evidence
1	Empty-domain subtraction	Extracted scattered coefficients remain below the solver and interpolation noise floor.
2	Analytic sphere benchmark	Numerical T-matrix reproduces Mie coefficients; symmetry-forbidden and off-diagonal entries converge toward zero.
3	Incident-matrix conditioning	$\mathrm{rank}(\mathbf{A}_w) = D$; singular spectrum and condition number are reported; results are stable to reasonable SVD-cutoff changes.
4	Projection and holdout	ϵ_{inc}, ϵ_{proj}, ϵ_{fit}, and ϵ_{hold} are reported; a recommended initial target is 10^{-3} or lower.
5	Numerical-domain convergence	Mesh refinement, PML distance/thickness, monitor radius, and surface quadrature changes alter the retained T-matrix and selected observables by less than the declared tolerance.
6	Multipole convergence	Increasing $L_{\max}$ by one or two does not materially change the common T block or observables.
7	Dimer benchmark	A two-particle T-matrix calculation agrees with a direct full-wave dimer simulation before larger arrays are attempted.
8	Finite or periodic benchmark	A small finite cluster agrees with a full-wave far-field result, or a periodic T-matrix calculation agrees with a Floquet full-wave unit-cell result.
9	Conservation and reciprocity	All open channels and material absorption satisfy power balance; reciprocity is checked using the permutation and phase convention of the serialized VSWF basis.

For the normalized regular/outgoing convention, $j_l = (h_l^{(1)} + h_l^{(2)})/2$ gives the incoming/outgoing scattering matrix

$$\mathbf{S} = \mathbf{I} + 2\mathbf{T}. \tag{36}$$

For a lossless meta-atom, $\mathbf{S}^\dagger\mathbf{S} = \mathbf{I}$. For a passive lossy meta-atom, $\mathbf{S}^\dagger\mathbf{S} \preceq \mathbf{I}$. Equivalently,

$$\mathbf{Q} = -\mathbf{T} - \mathbf{T}^\dagger - 2\mathbf{T}^\dagger\mathbf{T} \succeq 0. \tag{37}$$

If a differently scaled basis is used, insert its flux Gram matrix rather than applying these Euclidean forms unchanged [5].

8 Required metadata and file format

A bare array such as `T_Matrix.npy` is insufficient for reproducibility. Store the complex matrix in a self-describing HDF5 file, or provide a mandatory sidecar manifest containing:

- geometry identifier, material model identifiers, constitutive tensors, and frequency units;
- background $\epsilon_b, \mu_b, k_b, Z_b$ and time dependence $\exp(-i\omega t)$;
- expansion origin, object orientation, length units, and explicit ordered mode list;
- definitions of $Y_{lm}, \mathbf{M}_{lm}^{(p)}, \mathbf{N}_{lm}^{(p)}$ and radial-function conventions;
- $L_{\max}$, monitor radii, surface quadrature, interpolation method, and projection residuals;
- every incident direction, polarization, amplitude, weight, and incident coefficient vector;
- mesh/adaptation settings, PML distance/thickness/grading, solver version, and code commit;
- SVD cutoff, singular values, effective rank, fit error, holdout error, and validation status.

The recommended stored matrix shape is (N_f, D, D), with a separate frequency vector and no silent reordering of modes. A shared convention and self-describing format are essential when T-matrices will be reused across solvers or machine-learning pipelines [9].

9 Recommended execution order

The first implementation should proceed in the following order:

1. Validate the projection and normalization with an analytic dielectric or metallic sphere.
2. Extract one realistic-material meta-atom using the full-sphere excitation set.
3. Pass independent holdout illuminations and all PML, mesh, radius, quadrature, and $L_{\max}$ convergence tests.
4. Validate a homogeneous-medium dimer against a direct full-wave simulation.
5. Validate a small finite cluster. Add the periodic Bloch path only if an infinite repeated surface is the intended physical problem.
6. Introduce symmetry reduction or large-system acceleration only after the unreduced reference calculation is stable.

This staged procedure isolates numerical extraction errors from multiple-scattering errors and provides a physically interpretable baseline before additional environments are considered.

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